Phase 4 - Knowledge Definition

6 Knowledge Definition

The goal of the Knowledge Definition phase is to transform the informal semantic structures described in the Language Teleontology into a formalized, machine-interpretable Knowledge Teleontology. In this phase, the focus is on refining the conceptual model by assigning appropriate domains and ranges, formalizing ETypes, and defining object and data properties in accordance with the ITELOS methodology.

To achieve this, we systematically evolved the language-level model into a knowledge-level ontology by enriching each element with semantic constraints, explicit IRIs, and logical axioms. This ensures that the ontology is both syntactically correct and semantically coherent, enabling downstream reasoning and interoperability with external datasets.

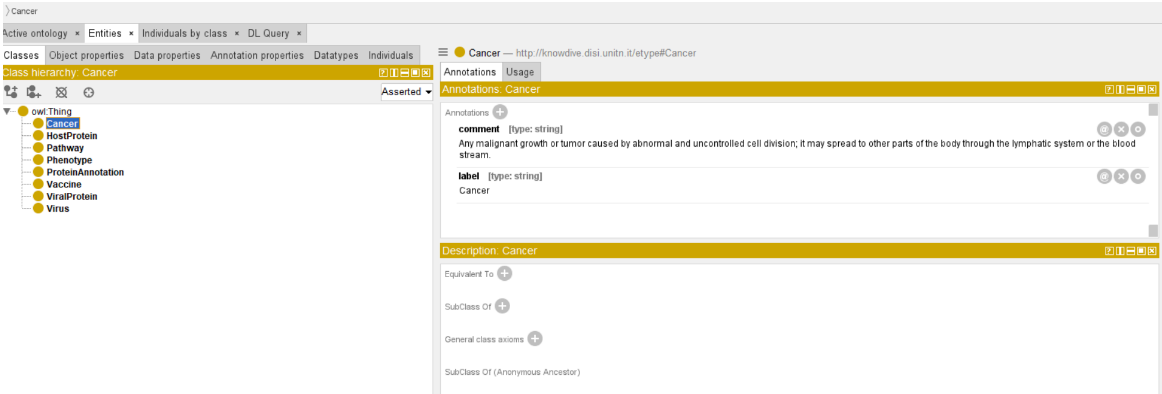
6.1 EType Formalization

We began the formalization process by introducing a root EType, named “Thing”, acting as the universal superclass for all ETypes in the ontology. This provides a consistent hierarchical structure and ensures compatibility with standard ontology engineering practices.

For each EType identified in the informal ER/EER model, we performed the following steps:

- Created a new subclass of Thing representing the domain entity. No additional subclass hierarchies were required for this project, so all ETypes remain direct descendants of Thing.
- Updated the IRI of each class to follow the required naming format: http://knowdive.disi.unitn.it/etype#<EType_name>
- Added annotations to each EType, including:
    - Label
    - Definition
    - Source of the information, referencing the Language Resource Table (Table 3)

This ensures that each EType is uniquely defined, traceable to its data origin, and aligned with the semantic constraints of the knowledge layer.

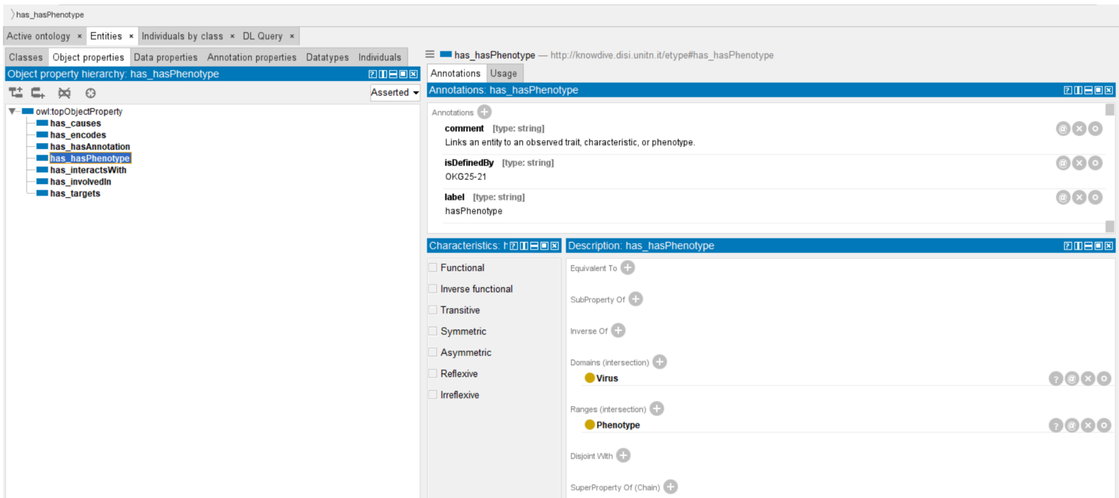


6.2 Object properties Formalization

Object properties formalize the relationships between ETypes and act as the structural backbone of the ontology. For each relationship defined in the language teleontology, we executed the following steps:

- Created a new object property named according to the language-level definition.
- Annotated the property with its definition and Concept ID.
- Updated its IRI using the required pattern: http://knowdive.disi.unitn.it/etype#has_<property-name>
- Specified the Domain(s) and Range(s) based on the ETypes involved in the relationship.
  This step formalizes the directional semantics—i.e., indicating the class the relationship originates from and the class it points to.

This process enabled the representation of inter-entity relationships in a structured and machine-interpretable form.

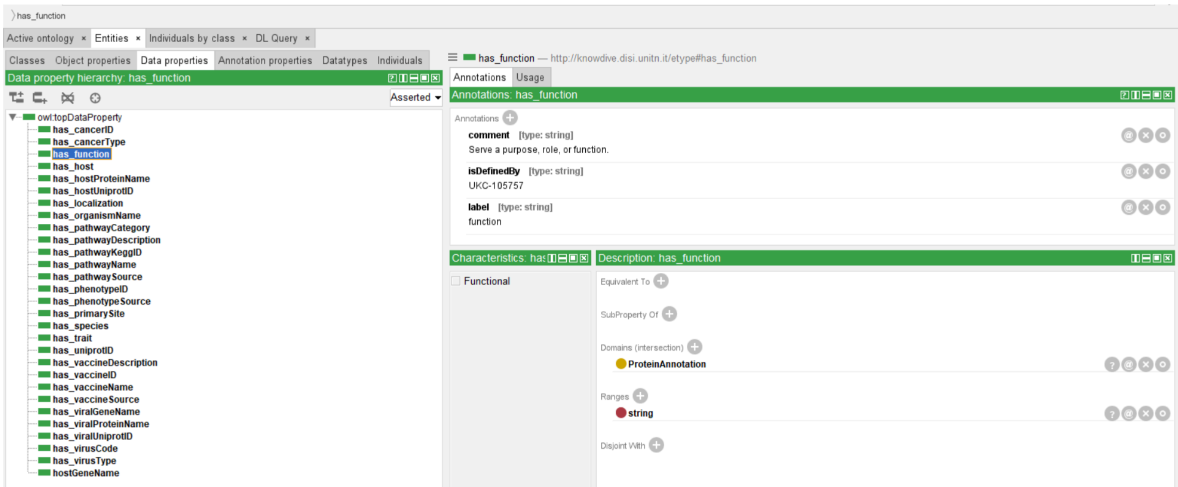


6.3 Data properties Formalization

Data properties capture the literal attributes of each EType. For every data property extracted from the informal ER/EER model, we applied the following procedure:

- Created a new data property named according to the corresponding language-level label.
- Annotated it with its definition and Concept ID.
- Updated its IRI to follow the standardized format:
  http://knowdive.disi.unitn.it/etype#has_<property-name>
- Assigned Domain(s) corresponding to the EType that owns the attribute.
- Assigned Range(s) to reflect the datatype of the attribute (e.g., Boolean, String, Float).

This step ensures that each attribute is precisely defined and constrained, enabling validation and reasoning across instances.



6.4 Teleontology

Because this project focuses on a highly specialized biomedical domain, no existing teleontology fully matched our needs. Therefore, we constructed a custom teleontology using multiple ontologies available through the NCBI BioPortal.

Since we searched for exact term matches across BioPortal resources, we did not require ontology-to-ontology alignment procedures. However, the resulting teleontology still contains some redundancy; thus, instead of detailing each individual class and property, we provide a conceptual overview of the structure below.

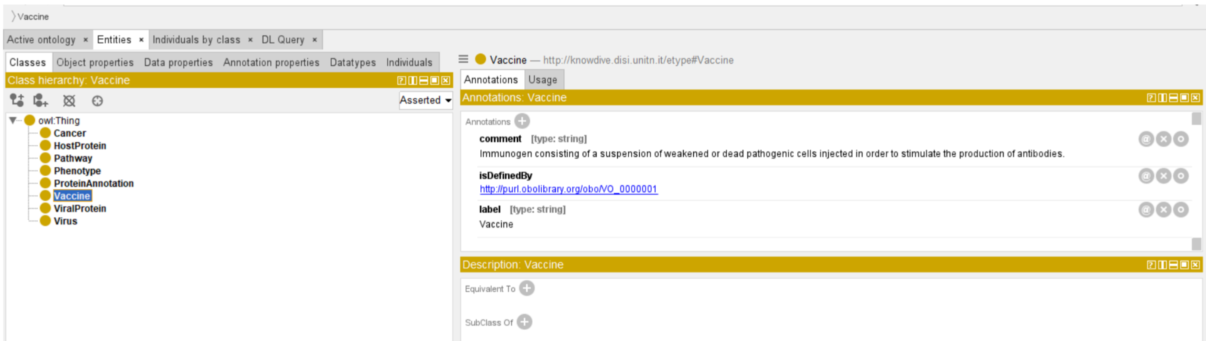
6.4.1 Etype

Each EType was aligned with the most appropriate external ontology. Examples include:

- Cancer aligned using the Human Disease Ontology

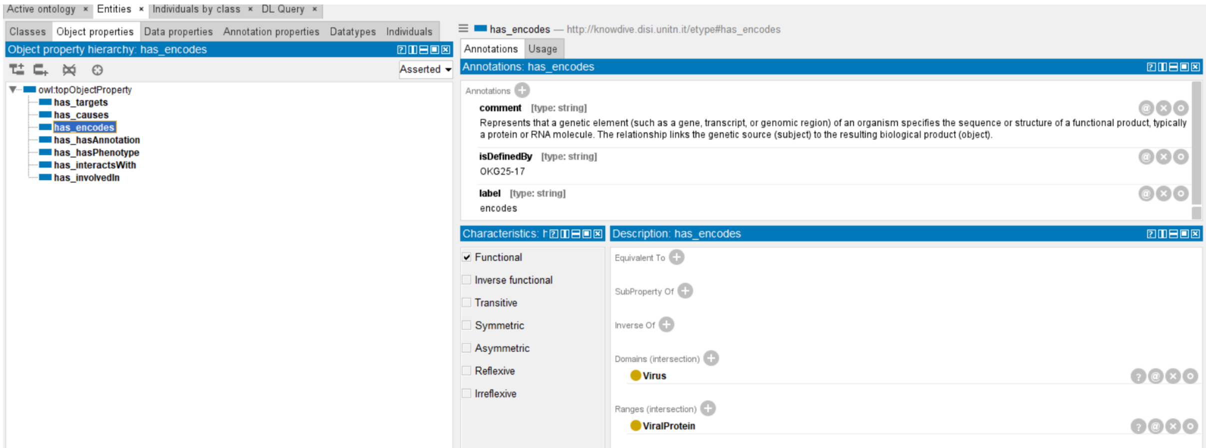
- Virus aligned using the Semantic Types Ontology
- Vaccine aligned using the Vaccine Ontology

These alignments ensured consistency of terminology and compatibility with established biomedical knowledge sources.



6.4.2 Object properties

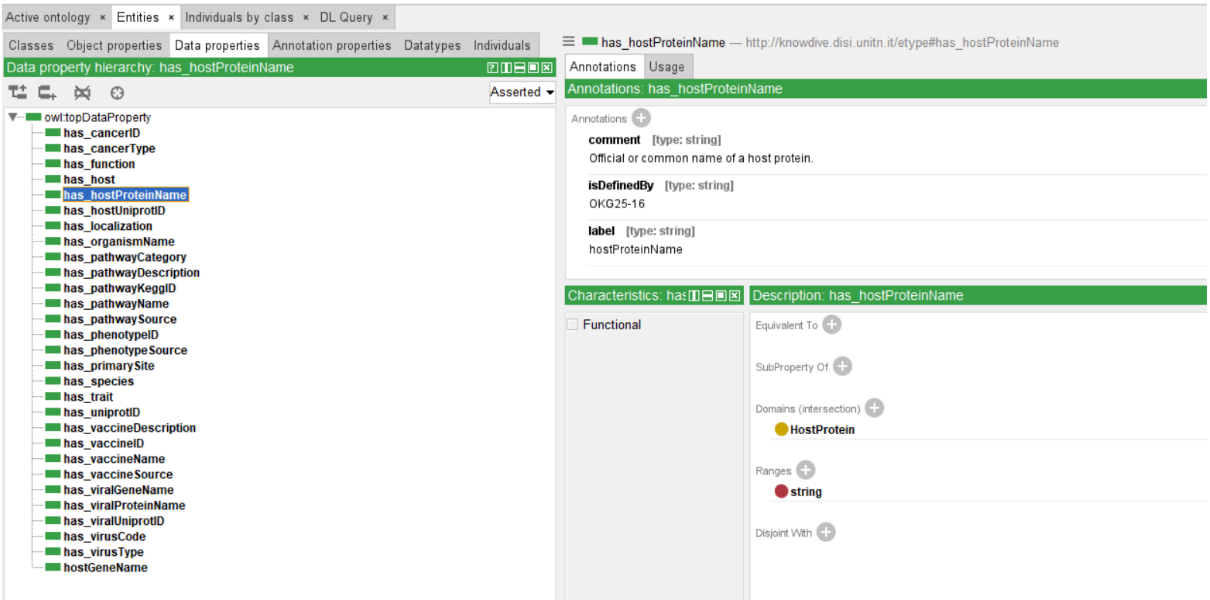
For object properties, we were generally unable to locate exact matches within existing ontologies. As a result, most object properties were defined internally, based on the semantic requirements of the project.



6.4.3 Data properties

The external ontologies contained numerous attributes and branches that extended beyond the scope of our project. To maintain clarity and relevance:

- Only the relevant attributes were reused.
- A majority of data properties were defined manually, tailored specifically to our use case.



Our final teleontology (depicted in the referenced figure) contains eight ETypes (classes), along with eight object properties and twenty-nine data properties (counted as 8 and thirty due to an extra "intuition" header in these cases). The ontology also includes four annotation properties, seventy-eight logical axioms, and a total of 261 axioms overall. The complete ontology was exported in OWL (.rdf) format and uploaded to GitHub under the name oncovirus_KG_knowledge_teleontology.owl.

Ontology metrics:

Metrics

Axiom	260
Logical axiom count	77
Declaration axioms count	51
Class count	8
Object property count	7
Data property count	30
Individual count	0
Annotation Property count	3

6.5 Qualitative Evaluation

During this phase, several refinements were introduced to improve consistency, integration, and compliance with the project guidelines. The virus–cancer lookup table was removed and replaced with an explicit attribute within the virus dataset that directly specifies the associated cancer type, resulting in a more streamlined representation of this relationship. To strengthen dataset interoperability, the viral protein dataset was enriched with interaction IDs and host UniProt IDs, allowing interaction information to be incorporated directly rather than through pivot or lookup tables. Additionally, the IRIs within the language teleontology OWL file were revised to conform to the naming conventions required by the guideline sheet, ensuring semantic and structural uniformity. Finally, all object and data properties in the teleontology were updated so that their names begin with the prefix “has_” aligning them with the standard property-naming format prescribed by the ITELOS methodology.